

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                                                                                                               | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(a)            | <p>A description that includes five of the following points:</p> <ul style="list-style-type: none"> <li>• dependent variable identified (1)</li> <li>• range of at least five suitable temperatures (1)</li> <li>• description of how to obtain quantitative results (1)</li> <li>• time measurement to obtain rate (1)</li> <li>• carbon dioxide needs to be absorbed (1)</li> <li>• control of temperature with a thermostatic water bath (1)</li> <li>• {same / stated} time for exposure to each temperature to equilibrate (1)</li> <li>• repeats (at each temperature) and calculate a {mean / standard deviation} (1)</li> </ul> | <p>For example, oxygen consumption</p> <p>Accept temperatures within the range of 0 to 40 °C, e.g. measurement of coloured liquid movement / use of a respirometer / use of hydrogen carbonate indicator</p> <p>Accept reference to use of KOH</p> <p>Accept description of how temperature is controlled, e.g. Bunsen burner and thermometer</p> | (5)  |

| Question number | Answer                                                                                                                                                                                                                                                    | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)            | <p>An answer that includes any two of the following:</p> <ul style="list-style-type: none"> <li>• age of seeds (1)</li> <li>• {species / variety} of seeds (1)</li> <li>• {mass / number} of seeds (1)</li> <li>• water available to seeds (1)</li> </ul> | (2)  |

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|-----------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------|
| 1(c)(i)         | <ul style="list-style-type: none"> <li>• variable with suitable control method described</li> </ul> | For example, (age of seeds) choose seeds from the same plant / pod / packet | (1)  |

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|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------|
| 1(c)(ii)        | <ul style="list-style-type: none"> <li>• results are not valid / description of expected effect on the dependent variable</li> </ul> | For example, older seeds may respire more slowly | (1)  |

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|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------|
| 1(d)            | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• reduced growth because {increase in anaerobic / decrease in aerobic} respiration (1)</li> <li>• therefore less ATP produced (1)</li> <li>• therefore less energy available for growth (1)</li> </ul> | <p>Accept ethanol produced</p> <p>Accept (ethanol)</p> | (3)  |